

Phase 6 – Sales Trend Analysis

1. Which year generated the highest sales?

Answer: 2021 generated the highest sales — 6,039,494.

2. Which months have the highest sales?

Answer: January had the highest sales — 1,515,931.

3. Which months have the lowest sales?

Answer: August had the lowest sales — 931,854.

4. Is the business growing year over year?

Answer: No. Sales decreased from 6,039,494 in 2021 to 5,951,226 in 2022, a decline of about 1.46%.

5. Are there any unusual sales spikes or drops?

Answer: Yes. January shows a strong sales peak, while August shows a noticeable low point.

6. Which quarter performs best?

Answer: Q1 performs best, with sales of 4,239,822.

7. How does the current year's performance compare with the previous year?

Answer: 2023 sales are currently lower than 2022. However, 2023 contains only January–March data, so a full-year comparison is not fair. For Jan–Mar, 2023 is about 14.8% lower than Jan–Mar 2022.

Phase 7 – Product Performance Analysis

1. Which products generate the highest revenue?

Answer: Product 47 generates the highest revenue — 296,430.

2. Which products generate the highest profit?

Answer: Product 41 generates the highest profit — 25,388.

3. Which products have high sales but low margins?

Answer: Product 63 is a major concern. It has high sales (280,896) but a very low margin of about -61.0%. Product 84 is another example (-60.2% margin).

4. Which products have the highest quantity sold?

Answer: Product 24 has the highest quantity sold — 289 units.

5. Which products should management prioritize?

Answer: Management should prioritize products that have high revenue, strong profit, and healthy margins, especially Product 41, which has the highest profit.

6. Which products may require pricing or cost optimization?

Answer: Products such as Product 63 and Product 84 require pricing or cost optimization because they generate high sales but have very negative margins.

Phase 8 – Customer Analysis

1. Who are the top customers?

Answer: The top customer by revenue is Aaron Miller, with sales of 34,647.

2. What percentage of revenue comes from the top 10 customers?

Answer: The top 10 customers contribute approximately 2.4% of total revenue.

3. Which customers generate the highest order volume?

Answer: Aaron Miller has the highest order volume, with 27 orders.

4. Which customers contribute the highest profit?

Answer: Phillip Morrison contributes the highest profit, with 5,788.

5. Are there customers with high sales but low profitability?

Answer: Yes. For example, Mark Morales has sales of 30,903 but a negative profit of -16,156, indicating low profitability despite high sales.

6. Which customers should the organization prioritize?

Answer: Management should prioritize high-sales and high-profit customers, while reviewing high-sales customers with negative profit to improve pricing, discounts, or costs.

Phase 9 – Salesperson Performance

1. Who is the highest-performing salesperson?

Answer: Ryan Welch is the highest-performing salesperson by sales, generating 344,324.

2. Who generates the highest profit?

Answer: Sean Miller generates the highest profit at -55,072.

Profit is negative overall, so this is the salesperson with the least negative profit.

3. Who handles the highest number of orders?

Answer: Ryan Welch handles the highest number of orders — 273 orders.

4. Which salespeople are improving year over year?

Answer: Several salespeople improved from 2021 to 2022. Carl Hall showed the highest improvement, increasing sales by about 21.9%.

5. Are there salespeople with high sales but low profitability?

Answer: Yes. Ryan Welch has the highest sales (344,324) but a negative profit of -102,342, showing high sales but poor profitability.

6. Which salespeople may require additional support or training?

Answer: Salespeople with high sales but negative/low profit, such as Ryan Welch and Kenneth Bradley, may need support in pricing, discount management, and cost control.

Phase 10 – Geographic Analysis

1. Which states generate the highest revenue?

Answer: California generates the highest revenue, with total sales of 13,254,991.

2. Which locations are performing best?

Answer: Rialto performs best, generating 227,227 in sales.

3. Which locations have low sales?

Answer: San Francisco has relatively low sales at 147,668, followed by Murrieta at 148,642.

4. Does population appear to influence sales?

Answer: No strong relationship is visible. Population and sales have almost no correlation, so higher population does not necessarily mean higher sales.

5. Are there locations with high population but relatively low sales?

Answer: Yes. San Francisco has a high population (864,816) but relatively low sales (147,668), indicating potential underperformance.

6. Which locations could represent potential growth opportunities?

Answer: Locations with high population but relatively low sales, such as San Francisco and Fresno, could represent potential growth opportunities

Phase 11 – Budget vs Actual Analysis

1. Which locations exceeded their budget?

Answer: 74 locations exceeded their budget. Rialto was one of the strongest performers, with actual sales of 227,227 against a budget of 44,500.

2. Which locations failed to achieve their budget?

Answer: 23 locations failed to achieve their budget. Some locations recorded no sales, resulting in 0% budget achievement.

3. What is the overall budget achievement percentage?

Answer: Overall budget achievement is approximately 313.0%.

4. Which months have the highest budget variance?

Answer: January has the highest positive budget variance, at approximately 1,346,547.

5. Which locations consistently underperform against budget?

Answer: Locations with zero or very low sales compared with their budget, such as A180, A181 and A182, are consistent underperformers.

6. Which locations should management investigate?

Answer: Management should investigate locations with very low or zero sales against their budget, especially A180, A181 and A182, to identify the reasons for underperformance.

Management Questions – Final Challenge

Business Performance

1. What is the total revenue?

Answer: Total revenue is 13,254,991.

2. What is the total profit?

Answer: Total profit is -4,062,325.

3. What is the overall profit margin?

Answer: Overall profit margin is approximately -30.6%.

4. Is revenue growing year over year?

Answer: No. Revenue decreased from 6,039,494 in 2021 to 5,951,226 in 2022.

5. Which year performed best?

Answer: 2021 performed best, with revenue of 6,039,494.

Products

6. Which are the top 10 products by revenue?

Answer:

1. Product 47 – 296,430
2. Product 28 – 294,646

3. Product 63 – 280,896
4. Product 84 – 274,048
5. Product 29 – 267,111
6. Product 56 – 259,480
7. Product 59 – 250,527
8. Product 66 – 249,184
9. Product 1 – 244,269
10. Product 41 – 239,903

7. Which are the top 10 products by profit?

Answer:

1. Product 41 – 25,388
2. Product 77 – 13,966
3. Product 96 – 10,356
4. Product 2 – 5,940
5. Product 26 – 3,389
6. Product 76 – 2,352
7. Product 45 – 2,070
8. Product 98 – 2,040
9. Product 91 – 576
10. Product 61 – -399

8. Which products have poor margins?

Answer: Product 63 and Product 84 have particularly poor margins because they generate high revenue but significant negative profit.

9. Which products have the highest demand?

Answer: Product 23 has the highest demand, with 289 units sold.

Customers

10. Who are the top 10 customers?

Answer: The top customer is Aaron Miller, followed by Russell Boyd, Jeffrey Sanders, Philip Burton, William Medina, Louis McCoy, Mark Morales, Jeremy Arnold, Fred Russell, and Victor Gray.

11. What percentage of sales comes from the top customers?

Answer: The top 10 customers contribute approximately 2.4% of total sales.

12. Which customers generate the highest profit?

Answer: Phillip Morrison generates the highest profit, at 5,788.

Sales Team

13. Who is the best-performing salesperson?

Answer: Ryan Welch – highest sales at 344,324.

14. Who generated the highest profit?

Answer: Sean Miller – highest profit at -55,072 (*least negative profit in the current calculation*).

15. Which salesperson has the highest order volume?

Answer: Ryan Welch – 273 orders.

Geography

16. Which state generates the highest revenue?

Answer: California – 13,254,991 revenue.

17. Which locations are underperforming?

Answer: San Francisco and Murrieta are among the lowest-sales locations, with 147,668 and 148,642 respectively.

18. Which locations appear to have growth potential?

Answer: San Francisco and Fresno show growth potential because they have relatively high populations but comparatively low sales.

Budget

19. Is the organization achieving its budget?

Answer: Yes, according to the current dashboard calculation. Overall budget achievement is 313.0%.

20. Which locations exceeded the budget?

Answer: 74 locations exceeded their budget. Examples include A116, A146, A102, A149 and A160.

21. Which locations missed the budget?

Answer: 23 locations missed their budget. Examples include A180, A181 and A182, which recorded zero actual sales against their budgets.

22. Which months had the largest budget variance?

Answer: January had the largest positive variance, approximately 1,346,547, followed by February at approximately 1,163,885.

Strategic Insights

23. What are the three biggest strengths of the business?

Answer:

1. Strong revenue base – Total revenue is 13.25M.
2. Strong-performing products – Products such as Product 41 generate relatively high profit.
3. Strong locations – Rialto and several other locations generate strong sales and exceed their budget.

24. What are the three biggest areas of concern?

Answer:

1. Revenue is not growing consistently – 2022 revenue was lower than 2021.
2. Negative profitability – Overall profit and profit margin are negative.

3. Underperforming products and locations – Products such as Product 63 and Product 84 have very poor margins, while some locations have very low or zero sales.

25. What recommendations would you give management based on your analysis?

Answer:

1. Improve pricing and cost control for products with negative/low margins.
2. Focus on high-performing products and locations and replicate their successful strategies.
3. Investigate underperforming locations and customers and create targeted sales strategies.
4. Monitor revenue trends closely and identify reasons for year-over-year decline.
5. Review the budget allocation so budgets are aligned correctly with the corresponding year and location.